# Sahaj Bhadja

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## Summary

Forward Deployed Software Engineer with 2+ years translating ambiguous business workflows into **AI-assisted applications, internal platforms, cloud services, and data products** through rapid prototyping and embedded stakeholder collaboration.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Advanced NLP, Analysis of Algorithms

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

## Experience

**USC Dornsife** **September 2025 – Present**
*Forward Deployed Software Engineer* *Los Angeles, CA*

- Embedded with stakeholders to translate operational workflows into FastAPI services and dashboards, processing 1K+ concurrent AI jobs at 3x throughput through **technical discovery, rapid prototyping, deployment, and iterative feedback**.
- Architected a multi-tenant Azure SQL platform for 500+ users, reducing dashboard latency 71% through **data modeling, indexed queries, row-level security, and end-to-end ownership from ambiguous requirements to production**.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Delivered React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **user interviews, prototype iteration, REST APIs, and production releases**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **workflow automation, testing, debugging, and measurable optimization**.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* *India*

- Designed cross-platform APIs and data pipelines using object-oriented patterns, AWS Glue, S3, and caching, improving response times 30% while reducing technical debt 20% across distributed business-critical services.
- Automated 20+ unit and integration suites across Windows, Linux, and macOS using PyTest, Mockito, Jenkins, and virtual machines, strengthening **CI/CD, release validation, technical decision-making, and cross-team delivery**.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern - (VibrantChem)* *India*

- Built React and Node.js services for resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, reusable TypeScript components, and role-aware internal operational workflows** across locations.
- Automated Docker and Jenkins delivery, reducing deployment time 50% while using Sentry, Azure Monitor, and Kibana for **observability, user-impact investigation, and reliable weekly product releases** across environments.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*

- Led discovery and presented solution architecture for 12+ integrations, mapping booking, payment, coupon, and status workflows to **align technical and non-technical stakeholders while reducing design errors 25%**.
- Built Flask APIs supporting 50K+ records with sub-200-millisecond responses while delivering **interactive demos, implementation guidance, customer feedback loops, and ownership through launch** for booking and payment workflows.

## Skills

**Languages & Web:** Python, TypeScript, JavaScript, Java, C++, React, Next.js, Flask, FastAPI, HTML5, CSS3
**AI & Product:** AI Agents, Claude Code, LLMs, RAG, Prompt Engineering, Rapid Prototyping, Product Discovery, User Feedback
**Data & Backend:** REST APIs, WebSockets, PostgreSQL, Azure SQL, MySQL, Redis, Data Modeling, Storage Systems, Event-Driven Systems
**Cloud & Infrastructure:** AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Monitoring, Observability
**Delivery:** Solution Architecture, Requirements Scoping, Internal Tools, Technical Demos, Agile, Testing, Debugging, Stakeholder Communication, Ownership

## Projects

**PropertyOps Copilot** | *Python, TypeScript, React, LLMs, PostgreSQL, Azure*

- Designed an **AI-assisted property operations workspace** that converts maintenance requests, tenant feedback, and asset records into prioritized actions, grounded recommendations, approval workflows, and auditable follow-ups for facilities teams.

**Asset Intelligence Workbench** | *React, FastAPI, PostgreSQL, GeoJSON, AWS*

- Built a **healthcare real-estate decision platform** combining property performance, lease events, service issues, and geospatial context into interactive dashboards, scenario comparisons, automated alerts, and stakeholder-ready operational recommendations.

**Big-Brain** | *LLaMA 2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **retrieval-augmented AI knowledge platform** indexing 50K+ documents with LLaMA 2, FAISS, Pinecone, PostgreSQL, and REST APIs to deliver grounded answers through a deployed GCP-backed application.